# Linear Algebra & Applications

MTH10013



Hello and welcome

Each week

- 2 x 2-hour Lecture
- 1 x 1-hour Tutorial
- 1 x Online assignment

Assessments

- 12 x Weekly assignments
- Test week 6 (in lecture Monday)
- Test week 11 (in lecture Monday)

BOTH TESTS ON CAMPUS FOR EVERYONE

- Exam (with hurdle)

Support

- Ask me questions

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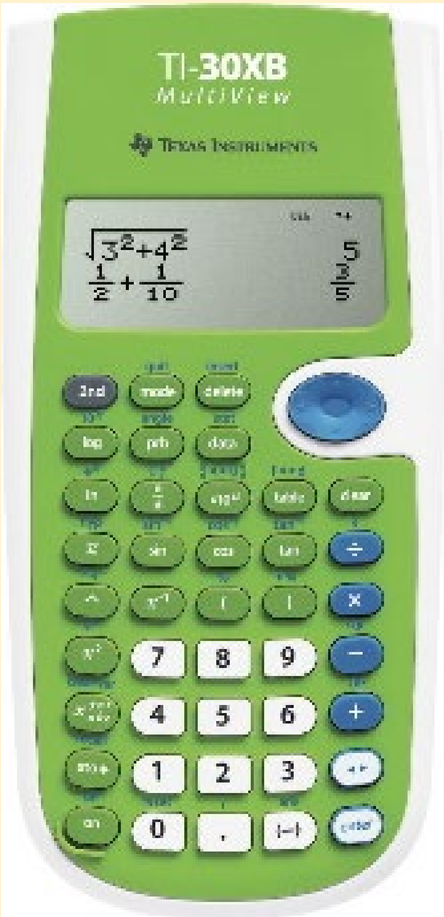
# Introductions



# Swinburne Mathematics and Statistics Help (MASH) Centre

- ✓ Drop-in support for individuals and groups
- ✓ Free, no appointment needed, just turn up
- ✓ Help with your first-year Maths units

MASH CENTRE

www.swinburne.edu.au/mash/

mash@swin.edu.au

# Location

- We are located in room AMDC503

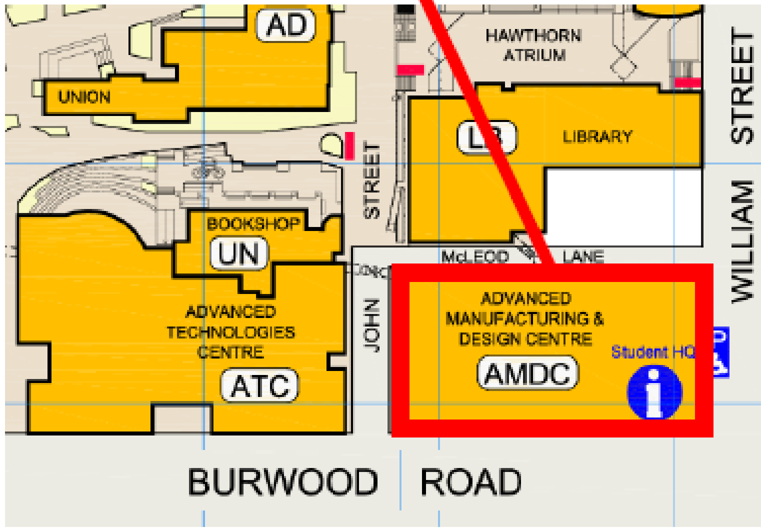
  *Mondays to Fridays, 10:30 – 16:30 during semester*

  *We have also some hours of online support in Collaborate Ultra*

  *(see timetable in our Canvas shell)*

AMDC503 (Level 5
Monday to Friday
10:30am to 16:30pm

You can self-enrol at the MASH Canvas shell to see the timetables for specialist and online support times.
Go to: [https://swinburne.instructure.com/enroll/9MJB7R](https://swinburne.instructure.com/enroll/9MJB7R)
or scan the QR code.




| | 10:30-11:30 | 11:30-12:30 | 12:30-1:30 | 1:30-2:30 |
|---|---|---|---|---|
| Monday | Sanka | | | |
| | | Meysam | Emma [ML] **(online)** | |

| |
|---|
| physics |
| second year maths |
| statistics |
| [ML] MatLab |

# Why visit MASH?



- ✓ Want to go over your lecture notes
- ✓ Want to improve your grades
- ✓ Need help with preparing for exams
- ✓ Have not studied maths recently
- ✓ Studied maths overseas and are not sure if you have the required background
- ✓ Lack confidence in your mathematical abilities
- ✓ Need some help with your calculator
- ✓ Just want to find a quiet space to study maths, knowing there is help nearby if you get stuck.

www.swinburne.edu.au/mash/

email: mash@swin.edu.au

# Unit Outline and Canvas Structure

# 1.1 What is a matrix?

Definition
Notation
Transpose

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# What is a matrix?

**A matrix is an array of numbers (elements)**

**Elements are arranged in *m rows* x *n columns***

**The Order of the matrix is m x n**

$$A=\begin{bmatrix}1 & 3\\ 5 & 2\end{bmatrix}$$

$$B=\begin{bmatrix}1 & 2 & -5\\ 0 & 3 & 2\\ -2 & 5 & 0\end{bmatrix}$$

$$C=\begin{bmatrix}1\\ 5\\ -2\end{bmatrix}$$

$$D=\begin{bmatrix}4.4 & -3.2 & 0\\ 17.5 & 3 & 11.5\end{bmatrix}$$

$$E=\begin{bmatrix}3 & 7 & -1\end{bmatrix}$$

# Applications – solving systems of equations

***Solve:***

$$2x - y = 1$$
$$x + y = 8$$

***Solve:***

$$5a - b + 3c - d + 7e = 11$$
$$a + b + c + d + e = 1$$
$$a - b + c - d + e = -1$$
$$3a - 2c + e = 14$$
$$101a - 8b + c - 2d + e = 0$$

**When looked at with matrices, they're the same thing…**

# Notation

**A general matrix with m rows and n columns:**

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$A=\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

The elements of the matrix are denoted by double subscripted small letters: $a_{ij}$ , where i is the row number and j is the column number.
So the element $a_{34}$ is located in the 3rd row and 4th column.

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# Elements

$$A=\begin{bmatrix} 1 & 3 \\ 5 & 2 \end{bmatrix} \quad B=\begin{bmatrix} 1 & 2 & -5 \\ 0 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix}$$

$a_{21}$

$a_{12}$

$b_{32}$

# Transpose

**Swap rows with columns**

$$B = \begin{bmatrix} 1 & 2 & 6 \\ 0 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 3 & 5 \\ 6 & 2 & 0 \end{bmatrix}$$

# Write the transpose of:

$$C=\begin{bmatrix}1\\5\\-2\end{bmatrix} \qquad C^T=\begin{bmatrix}1 & 5 & -2\end{bmatrix}$$

$$A=\begin{bmatrix}1 & 3\\5 & 2\end{bmatrix} \qquad A^T=\begin{bmatrix}1 & 5\\3 & 2\end{bmatrix}$$

$$D=\begin{bmatrix}4.4 & -3.2 & 0\\17.5 & 3 & 11.5\end{bmatrix} \qquad D^T=\begin{bmatrix}4.4 & 17.5\\-3.2 & 3\\0 & 11.5\end{bmatrix}$$

# Now try the questions
# Either on Canvas (Exercise 1.1)
# or Study Guide pg 3

# 1.2 Matrix Operations

Addition/Subtraction
Multiplying a matrix by a scalar
Multiplying a matrix by a matrix

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# Addition/ Subtraction

**Can only add/subtract matrices if they are of the same order.**

**Add/subtract corresponding elements**

$$A=\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \qquad B=\begin{bmatrix} -5 & 12 \\ 1 & -7 \end{bmatrix}$$

**A+B=**

**A-B=**

**Commutative: A + B = B + A**

**Associative: (A+B) + C = A + (B+C)**

# Multiplication – 1. By a scalar

**Every element of the matrix is multiplied by a scalar (number) constant.**

**E.g.**

$$B = \begin{bmatrix} 1 & 2 & 6 \\ 0 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix}$$

$$3B = \begin{bmatrix} 3 & 6 & 18 \\ 0 & 9 & 6 \\ -6 & 15 & 0 \end{bmatrix}$$

# Factorisation

**Can take a common factor out of every matrix element**

$$\boldsymbol{C} = \begin{bmatrix} \mathbf{12} & \mathbf{-6} \\ \mathbf{3} & \mathbf{18} \end{bmatrix}$$

$$\boldsymbol{C} = \mathbf{3}\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

$$\boldsymbol{C} = \mathbf{4}\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

# Multiplication 2. Two matrices

**Multiply rows by columns and sum**

$$A = \begin{bmatrix} 5 & 7 & 9 \end{bmatrix} \text{ and B} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

**Find AB**

Ans=[92]

# Multiplication with Higher Orders

**Multiply rows by columns**

$$A = \begin{bmatrix} 5 & 7 & 9 \end{bmatrix} \text{ and B} = \begin{bmatrix} 2 & 0 \\ 4 & -3 \\ 6 & 1 \end{bmatrix}$$

**Find AB**

**Ans= [92   12] - 1 x 2 Matrix**

**Can you multiply BA?**

# Multiplication with Higher Orders

Given $A = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}$ and $\mathrm{B} = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, Find AB

# Questions

$$A = \begin{bmatrix} 3 & 3 \\ -2 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}$$

1. **Find A+B**

2. **Find A-3B**

3. **Find AB**

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 3 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & 2 \\ 2 & 0 & 0 \end{bmatrix}$$

1. **Find 2A+B**

2. **Find AB**

# Questions

$$A = \begin{bmatrix} 3 & 3 \\ -2 & 1 \end{bmatrix} \text{ and } \mathrm{B} = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}$$

1. **Find A+B**

2. **Find A-3B**

3. **Find AB**

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 3 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & 2 \\ 2 & 0 & 0 \end{bmatrix}$$

1. **Find 2A+B**

2. **Find AB**

# You can’t multiply all matrices!

**Generally, the product AB only exists if the adjacent dimensions of A and B are the same**

$$\underset{(m \times n)}{A} \quad \underset{(n \times p)}{B} \quad = \quad \underset{(m \times p)}{C}$$

# Multiplication Continued

**Multiply rows by columns**

$$A = \begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix} \text{ and } \mathrm{B} = \begin{bmatrix} 2 & 1 \\ 0 & -3 \end{bmatrix}$$

**Find AB**

**Can you multiply BA?**

**Find BA**

**AB not the same as BA**

**NOT Commutative**

# POLL QUESTION

$A=\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$ $B=\begin{bmatrix}2 & 0\\0 & 3\end{bmatrix}$ $C=\begin{bmatrix}2 & 0\\0 & 2\end{bmatrix}$ $D=\begin{bmatrix}0 & 2\\2 & 0\end{bmatrix}$

**Which of the following is/are true?**

Y. AB=BA
M. AC=CA
C. AD=DA
A. All of the above
X. None of the above

# LEGO Video - https://youtu.be/vQI-2XOCdAo

# 1.2 Matrix Operations 2

## Multiplying Multiple Matrices
## Transposition of a product

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# Multiplying Multiple Matrices

**Find CAB** $A = \begin{bmatrix} -1 & 4 \\ 6 & -3 \end{bmatrix}, \mathrm{B} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \, and \; \mathrm{C} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix}$

**(CA)B=**

# Multiplying Multiple Matrices

**Find CAB**

$$A = \begin{bmatrix} -1 & 4 \\ 6 & -3 \end{bmatrix}, \mathrm{B} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} and\ \mathrm{C} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix}$$

**C(AB)=**

**Associative**

**(CA)B=C(AB)**

**N.B. Not the same as ABC or BCA etc.  - NOT Commutative**

# Multiplying Transpose Matrices

$$\boxed{(AB)^T = B^T A^T}$$

**Showing this is true for** $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \mathrm{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$

# Now try the questions
# Either on canvas (self-test)
# or Study Guide pg 8

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# 1.3 Some special square matrices

Diagonal
Unit/Identity Matrix
Symmetric/ anti-symmetric
Null

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# Diagonal Matrices

**All elements not on the principal diagonal are zero**

$$D = \boxed{\begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix}}$$

**The product of any two diagonal matrices is commutative**

# Unit Matrices

**Also called Identity matrix (I)**

**Multiplication by other matrix doesn't change the other matrix**

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# Identity Matrix - examples

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -5 \\ 0 & 3 & 2 \\ -2 & 5 & 0 \end{bmatrix}$$

- **Find IC**

- **Find IB**

# Scalar Matrix

A diagonal matrix with all its main diagonal entries equal is a **scalar matrix**

$$D = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \lambda \mathrm{I}$$

# Triangular Matrices

**Upper-triangular**

**Elements below principal diagonal =0**

$$A = \begin{bmatrix} 2 & 3 & 2 \\ 0 & -4 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

**Lower-triangular**

**Elements above principal diagonal =0**

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 5 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

# Symmetric Matrices

A square matrix is <u>symmetric</u> if

$$A^T = A$$

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & -4 & 5 \\ 1 & 5 & 1 \end{bmatrix}$$

A square matrix is <u>anti-symmetric</u> if

$$B^T = -B$$

$$B = \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 3 \\ 5 & -3 & 0 \end{bmatrix}$$

# Null Matrix

**All elements are zero**

$$\boxed{A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$$

**A null matrix does not have to be square**

# Matrix Questions

**For the following matrices**

$$A=\begin{bmatrix} 2 & 3 \\ -4 & 2 \end{bmatrix}$$

$$B=\begin{bmatrix} -2 & 0 \\ 5 & -1 \\ 6 & 3 \end{bmatrix}$$

$$C=\begin{bmatrix} 4 & -3 & 0 \\ 7 & 3 & 1 \end{bmatrix}$$

$$D=\begin{bmatrix} 3 & -1 \\ 5 & 2 \end{bmatrix}$$

**Find the following or state why it doesn't exist: A+D, AD, BC, CB, BA, C-B$^T$**

# 1.4 Determinants

**Determinant**
**Matrix of co-factors**
**Adjugate (classical adjoint)**

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# What is a determinant?

**A number associated with a square matrix**

**Used for:**

- Finding Inverse Matrices
- Solving sets of linear equations
- Determining if vectors are linearly dependent

**Denoted by vertical lines**

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

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# Finding determinants

**For a 1 x 1 matrix**

$$A = [a_{11}]$$

$$det(A) = |a_{11}| = a_{11}$$

**For a 2 x 2 matrix**

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$det(A) = a_{11}a_{22} - a_{12}a_{21}$$

# Examples – Determinant of a 2 x 2 matrix

$A=\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$

$B=\begin{bmatrix} 4 & -2 \\ 6 & 2 \end{bmatrix}$

# POLL QUESTION

$$A=\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix} \qquad B=\begin{bmatrix}3 & 4\\1 & 2\end{bmatrix}$$

**How do the determinants of A and B relate to each other?**

A. **They are the same**
B. **They are reciprocals**
C. **They are negatives**
D. **They are not related**

# POLL QUESTION

$$A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B=\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

**How do the determinants of A and B relate to each other?**

A. They are the same
B. They are reciprocals
C. They are negatives
D. They are not related

# Determinant of a 3 x 3 (or higher) matrix

**For the ith column of A** $det(A) = \sum_{i,j=1}^{n} a_{ij} C_{ij}$

**Where C is the matrix of cofactors**

**In words… add the product of each element and its corresponding cofactor along a column or row.**

# Cofactor Matrix

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\begin{bmatrix} & & & \cdots \\ & & & \cdots \\ & & & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$\mathrm{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

# Cofactor Matrix

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$M_{ij}$ **- The minor of the element (the determinant of what is left if you delete row i and column j )**

$$\mathrm{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

# Cofactor Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

Ans=27

# Determinant – using row …?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

Ans=27

# Determinant – using column 1

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

Ans=27

# Adjugate (classical adjoint) Matrix

$$adj(A) = C^T$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} -3 & -39 & 36 \\ 6 & -3 & 0 \\ -3 & 15 & -9 \end{bmatrix}$$

$$C^T = \begin{bmatrix} -3 & & \\ -39 & & \\ 36 & & \end{bmatrix}$$

# Determinant Question

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 2 \\ -3 & 4 & 2 \end{bmatrix}$$

How to check your answer… excel/matlab

Ans=-30

# Which row/column would you pick to find det(A)?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 1 & 8 & 9 \end{bmatrix}$$

# Properties of the determinant

- $det(AB) = det(A)det(B)$ for any n x n matrices A and B
- $det\,(A^T) = det(A)$
- $de\,t(I) = 1$
- $\det(\lambda A) = \lambda^n \det(A)$ where A is an n x n matrix and $\boldsymbol{\lambda}$ is a scalar
- If matrix B is formed by swapping any two rows or columns of A then $det(B) = -det(A)$
- If matrix A contains two or more identical rows or columns $det(A) = 0$

# POLL QUESTION

$$A=\begin{bmatrix}1 & 2 & 3\\ 7 & 5 & 6\\ 4 & 8 & 9\end{bmatrix} \qquad B=\begin{bmatrix}1 & 3 & 2\\ 7 & 6 & 5\\ 4 & 9 & 8\end{bmatrix}$$

**How do the determinants of A and B relate to each other?**

A. **They are the same**
B. **They are reciprocals**
C. **They are inverses**
D. **They are negatives**
E. **They are not related**

# NOW TRY THE QUESTIONS EITHER ON CANVAS (EXERCISE 1.4) OR STUDY GUIDE PG 18

# 1.5 Matrix Inversion

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# Inverse Matrix

$\boldsymbol{A}^{-1}$ is the inverse of $\boldsymbol{A}$, where $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$ are n x n matrices

$$\boldsymbol{AX} = \boldsymbol{B}$$
$$\boldsymbol{A}^{-1}\boldsymbol{B} = \boldsymbol{X}$$

$$\boldsymbol{AA}^{-1} = \boldsymbol{I}$$
$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$$

e.g. given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, show\ that\ \ A^{-1} = \begin{bmatrix} -2 & 1 \\ {}^{3}/_{2} & {}^{-1}/_{2} \end{bmatrix}$

# Inverse of a 2 x 2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\boldsymbol{AA^{-1} = I}$$

# Inverse of a 2 x 2 Matrix

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and det(A)$\neq$ 0

$$\boxed{A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

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# Example

If $A = \begin{bmatrix} 1 & -2 \\ 4 & 2 \end{bmatrix}$, find $A^{-1}$

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# POLL QUESTION

$$A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$$

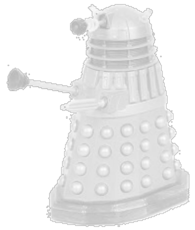
**Does the matrix A have an inverse $A^{-1}$?**

**A. Yes**
**B. No**
**C. Maybe**
**D. I don't know**
**E. Can you repeat the question?**

# Inverse Matrix (any n x n matrix)

$$\boxed{A^{-1} = \frac{1}{\det(A)} C^T(A)}$$

If det(A)=0 then $A^{-1}$ does not exist
A is said to be **singular**



Inverse matrices only exist for r x r matrices of rank r

# Inverse of a 2 x 2 matrix using

$$\boxed{A^{-1} = \frac{1}{\det(A)} C^T(A)}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

# Check Answer

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

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# Inverse of a 3 x 3 matrix

$$\boxed{A^{-1} = \frac{1}{\det(A)} C^T(A)}$$

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 4 & 1 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} \\ \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} \\ \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} & \begin{vmatrix} \quad & \quad \end{vmatrix} \end{bmatrix}$$

# Checking our answer...

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 4 & 1 & 2 \end{bmatrix}$$

# Questions

$$A = \begin{bmatrix} 3 & 3 \\ -2 & 1 \end{bmatrix}$$

Find $A^{-1}$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & 2 \\ 2 & 0 & 0 \end{bmatrix}$$

Find $B^{-1}$

$$A^{-1} = \frac{1}{\det(A)} C^T(A)$$

# More questions
# Either on canvas (Exercise 1.5)
# or Study Guide pg 22

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# 1.6 Matrix Algebra

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# For any n x n invertible matrices A and B:

- $If\ AB = I$ then $B = A^{-1}$ and $A = B^{-1}$
- $(A^{-1})^{-1} = A$
- $(kA)^{-1} = k^{-1}\ A^{-1}$ for an7 non-zero scalar k
- $(A^T)^{-1} = \ (A^{-1})^T$
- $\det(A^{-1}) = (\det(A))^{-1}$
- $(AB)^{-1} = B^{-1}\ A^{-1}$

**REMEMBER – Matrix multiplication NOT COMMUTATIVE**

# Inverse Matrix

$\boldsymbol{A^{-1}}$ is the inverse of $\boldsymbol{A}$, where $\boldsymbol{A}$ and $\boldsymbol{A^{-1}}$ are n x n matrices

$$\boldsymbol{AX = B}$$
$$\boldsymbol{A^{-1}B = X}$$

$$\boldsymbol{AA^{-1} = I}$$
$$\boldsymbol{A^{-1}A = I}$$

e.g. given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, show\ that\ \ A^{-1} = \begin{bmatrix} -2 & 1 \\ {}^{3}/_{2} & {}^{-1}/_{2} \end{bmatrix}$

# Example 1

Let A, B ,C and D be invertible matrices so that $BACD = I$.

Find an expression for A

# Example 2

Using the laws of matrix algebra show that

$$(\boldsymbol{ABC}^T)^{-1} = (\boldsymbol{C}^{-1})^T \boldsymbol{B}^{-1} \boldsymbol{A}^{-1}$$

# Questions

If AX=B, Find an expression for X.

If $(3BA)^{-1} = C^T$, find an expression for A in terms of B and C.